

Remark 1.30. We provided here, through Lemma 1.28, an explicit way of constructing a transport map between an atomless measure μ and an arbitrary measure ν , when the ambient space is $\mathbb{R}^d$. Actually, this fact is much more general, since it is well known that any measure space endowed with an atomless measure is isomorphic to $[0, 1]$ with the Lebesgue measure⁴. Yet, we do not want to introduce this kind of arguments when an explicit construction is sufficient to deal with the Euclidean case, which already meets our scopes.

A last lemma:

Lemma 1.31. *Consider on a compact metric space X , endowed with a probability $\rho \in \mathcal{P}(X)$, a sequence of partitions G_n , each G_n being a family of disjoint subsets $C_{i,n}$ such that $\bigcup_{i \in I_n} C_{i,n} = X$ for every n . Suppose that $\text{size}(G_n) := \max_i \text{diam}(C_{i,n})$ tends to 0 as $n \rightarrow \infty$ and consider a sequence of probability measures ρ_n on X such that, for every n and $i \in I_n$, we have $\rho_n(C_{i,n}) = \rho(C_{i,n})$. Then $\rho_n \rightarrow \rho$.*

Proof. Set $m_{i,n} := \rho_n(C_{i,n}) = \rho(C_{i,n})$. It is sufficient to take a continuous function $\phi \in C(X)$ and note that

$$\begin{aligned} \left| \int_X \phi d\rho_n - \int_X \phi d\rho \right| &\leq \sum_{i \in I_n} \left| \int_{C_{i,n}} \phi d\rho_n - \int_{C_{i,n}} \phi d\rho \right| \\ &\leq \omega(\text{diam}(C_{i,n})) \sum_{i \in I_n} m_{i,n} = \omega(\text{diam}(C_{i,n})) \rightarrow 0, \end{aligned}$$

where ω is the modulus of continuity of ϕ . This is justified by the fact that, whenever two measures have the same mass on a set $C \subset X$, since the oscillation of ϕ on the same set does not exceed $\omega(\text{diam}(C))$, the difference of the two integrals is no more than this number times the common mass.

This proves $\int \phi d\rho_n \rightarrow \int \phi d\rho$ and hence $\rho_n \rightarrow \rho$. □ □

We can now prove the following

Theorem 1.32. *On a compact subset $\Omega \subset \mathbb{R}^d$, the set of plans γ_T induced by a transport is dense in the set of plans $\Pi(\mu, \nu)$ whenever μ is atomless.*

Proof. Fix n , and consider any partition of Ω into sets $K_{i,n}$ of diameter smaller than $1/(2n)$ (for instance, small cubes). The sets $C_{i,j,n} := K_{i,n} \times K_{j,n}$ make a partition of $\Omega \times \Omega$ with size smaller than $1/n$.

Let us now take any measure $\gamma \in \Pi(\mu, \nu)$. Thanks to Lemma 1.31, we only need to build a transport T sending μ to ν , such that γ_T gives the same mass as γ to each one of the sets $C_{i,j,n}$. To do this, define the columns $\text{Col}_{i,n} := K_{i,n} \times \Omega$ and denote by $\gamma_{i,n}$ the restriction of γ on $\text{Col}_{i,n}$. Its marginal will be denoted by $\mu_{i,n}$ and $\nu_{i,n}$. Consider now, for each i , a transport map $T_{i,n}$ sending $\mu_{i,n}$ to $\nu_{i,n}$. It exists thanks to Corollary 1.29, since for each (i, n) we have $\mu_{i,n} \leq \mu$, which makes these measures atomless. Since the

⁴Note that this was the argument used in [176] to obtain the existence of transport maps in general spaces, and [253] points out that, when composing with the isomorphisms sending arbitrary spaces to $[0, 1]$, the cost function is no more continuous. Yet, this could be handled using Lemma 1.8, which allows to use costs of this form: the composition of a continuous function with measurable maps of x and y , separately.

$\mu_{i,n}$ are concentrated on disjoint sets, by “gluing” the transports $T_{i,n}$ we get a transport T sending μ to ν (using $\sum_i \mu_{i,n} = \mu$ and $\sum_i \nu_{i,n} = \nu$).

It is enough to check that γ_T gives the same mass⁵ as γ to every $C_{i,j,n}$, but it is easy to prove. Indeed, this mass equals that of $\gamma_{T_{i,n}}$ and $\gamma_{T_{i,n}}(C_{i,j,n}) = \mu_{i,n}(\{x : x \in K_{i,n}, T_{i,n}(x) \in K_{j,n}\}) = \mu_{i,n}(\{x : T_{i,n}(x) \in K_{j,n}\}) = \nu_{i,n}(K_{j,n}) = \gamma(K_{i,n} \times K_{j,n})$. $\square$ $\square$

The relaxation result is just a consequence.

Theorem 1.33. *Under the above-mentioned assumptions, K is the relaxation of J . In particular, $\inf J = \min K$, and hence Monge and Kantorovich problems have the same infimum.*

Proof. First note that, since K is continuous, then it is l.s.c. and since we have $K \leq J$, then K is necessarily smaller than the relaxation of J . We only need to prove that, for each γ , we can find a sequence of transports T_n such that $\gamma_{T_n} \rightarrow \gamma$ and $J(\gamma_{T_n}) \rightarrow K(\gamma)$, so that the infimum in the sequential characterization of the relaxed functional (see definition) will be smaller than K , thus proving the equality. Actually, since for $\gamma = \gamma_{T_n}$ the two functionals K and J coincide, and since K is continuous, we only need to produce a sequence T_n such that $\gamma_{T_n} \rightarrow \gamma$. This is possible thanks to Theorem 1.32 $\square$ $\square$

1.6 Convexity, c -concavity, cyclical monotonicity, duality and optimality

1.6.1 Convex and c -concave functions

In this section we analyze properties of c -concave functions in comparison with convex functions, which are better known. We start from recalling some notions from convex analysis.

Box 1.11. – Memo – Convex functions

The definition of convex function is always the same: $f : \mathbb{R}^d \rightarrow \mathbb{R} \cup \{+\infty\}$ is convex if and only if for all $x, y \in \mathbb{R}^d$ and $t \in [0, 1]$ we have $f((1-t)x + ty) \leq (1-t)f(x) + tf(y)$. We do not care about convex functions defined on subsets $\Omega \subset \mathbb{R}^d$ (which should be convex, for the definition to make sense) since one can always extend f outside Ω by setting it to $+\infty$, preserving convexity.

An easy stability property is the following: if f_α is a family of convex functions, then f defined through $f(x) := \sup_\alpha f_\alpha(x)$ is also convex. This can be checked easily by writing $f_\alpha((1-t)x + ty) \leq (1-t)f_\alpha(x) + tf_\alpha(y) \leq (1-t)f(x) + tf(y)$ and passing to the sup in α . Otherwise, we can also use the epigraphs (and a function is convex if and only if its epigraph is convex) and note that the interesection of convex sets is still convex.

⁵Note that [8] and [253] proved the same fact in a more complicated way: indeed, the use of a unique transport map for every column is enough. The same idea has been used in [147] for a different density result, concerning Lipschitz transport maps.

Real-valued convex functions defined on $\mathbb{R}^d$ are automatically continuous and locally Lipschitz (and in general, they are continuous and locally Lipschitz in the interior of their domain $\{f < +\infty\}$). However, on the boundary between $\{f < +\infty\}$ and $\{f = +\infty\}$ there could be discontinuities. This is why typically we require at least lower semi-continuity.

Theorem - A function $f : \mathbb{R}^d \rightarrow \mathbb{R} \cup \{+\infty\}$ is convex and l.s.c. if and only if there exists a family of affine functions f_α such that $f(x) := \sup_\alpha f_\alpha(x)$. This family can also be chosen to be the family of all affine functions smaller than f .

One implication is easy (if f is a sup of affine functions, since affine implies convex, then it is convex). The geometrical intuition behind the converse one is the fact that the epigraph of a convex and l.s.c. function is a closed convex set in $\mathbb{R}^d \times \mathbb{R}$, and can be written as the intersection of the half-spaces which contain it. A precise proof requires the use of Hahn-Banach Theorem in its geometrical form.

Box 1.12. – Memo – Legendre transform, subdifferential

Definition (Legendre-Fenchel transform) - for any given function $f : \mathbb{R}^d \rightarrow \mathbb{R} \cup \{+\infty\}$ we can define its transform $f^*(y) := \sup_x x \cdot y - f(x)$.

Example - the Legendre transform of $x \mapsto \frac{1}{p}|x|^p$ is $x \mapsto \frac{1}{q}|x|^q$, where $\frac{1}{p} + \frac{1}{q} = 1$.

A function f is convex and l.s.c. if and only if there exists g such that $f = g^*$ (since the class of functions that are Legendre transform of something exactly agrees with that of functions which are expressed as suprema of affine functions). To be convinced that every supremum of affine functions can be expressed as g^* , just take $f(y) = \sup_\alpha x_\alpha \cdot y + b_\alpha$; then, for each vector x , set $g(x) = -\sup\{b_\alpha : x_\alpha = x\}$ (by setting $g(x) = +\infty$ if no α is such that $x_\alpha = x$) and check that we have $f = g^*$.

Proposition - A function $f : \mathbb{R}^d \rightarrow \mathbb{R} \cup \{+\infty\}$ is convex and l.s.c. if and only if $f^{**} = f$.

The proof of this fact can be seen as an application of the Hahn-Banach theorem. For a complete proof, even in infinite dimension, see for instance [24].

Definition (subdifferential) - For every convex function f we define its subdifferential at x as the set $\partial f(x) = \{p \in \mathbb{R}^d : f(y) \geq f(x) + p \cdot (y - x) \forall y \in \mathbb{R}^d\}$.

It is possible to prove that $\partial f(x)$ is never empty if x lies in the interior of the set $\{f < +\infty\}$. At every point where the function f is differentiable, then ∂f reduces to the singleton $\{\nabla f\}$.

For the subdifferential of the convex functions f and f^* we have

$$p \in \partial f(x) \Leftrightarrow x \in \partial f^*(p) \Leftrightarrow f(x) + f^*(p) = x \cdot p.$$

This helps in proving the following equivalence: take two conjugate functions f and f^* (with $f = f^{**}$); then f is C^1 if and only if f^* is strictly convex. This comes from the fact that C^1 means that there is at most one p in every set $\partial f(x)$, while strictly convex means that the same vector cannot belong to the subdifferential of more than one point. In particular, if h is strictly convex then ∂h , which is a multi-valued map, can be inverted and is uni-valued, thus getting a map $(\partial h)^{-1}$, that should use in the statement of Theorem 1.17 instead of $(\nabla h)^{-1}$.

Finally, note that the subdifferential of a convex function satisfies this monotonicity property: if $p_1 \in \partial f(x_1)$ and $p_2 \in \partial f(x_2)$, then $(p_1 - p_2) \cdot (x_1 - x_2) \geq 0$.

Note the analogy with c -concave functions (or $\bar{c}$ -concave, this distinction being meaningless in the symmetric case), which were defined as being the $\bar{c}$ -transform (or c -transform) of something. Indeed, up to the change of sign, c -concavity has exactly been defined as a generalization of convexity. Note that we admit the value $+\infty$ for convex functions and $-\infty$ for concave and c -concave ones. Yet, we do not like the convex function which is identically $+\infty$; nor any c -concave function which is identically $-\infty$.

Instead of proving the result on $f^{**} = f$ we may prove this more general one concerning c -concave functions⁶.

Proposition 1.34. *Suppose that c is real valued. For any $\varphi : X \rightarrow \mathbb{R} \cup \{-\infty\}$ we have $\varphi^{c\bar{c}} \geq \varphi$. We have the equality $\varphi^{c\bar{c}} = \varphi$ if and only if φ is c -concave; in general, $\varphi^{c\bar{c}}$ is the smallest c -concave function larger than φ .*

Proof. First we prove $\varphi^{c\bar{c}} \geq \varphi$. Write

$$\varphi^{c\bar{c}}(x) = \inf_y c(x, y) - \varphi^c(y) = \inf_y c(x, y) - \inf_{x'} c(x', y) - \varphi(x').$$

Consider that $\inf_{x'} c(x', y) - \varphi(x') \leq c(x, y) - \varphi(x)$, and hence

$$\varphi^{c\bar{c}}(x) \geq \inf_y c(x, y) - c(x, y) + \varphi(x) = \varphi(x).$$

Analogously, we have $\zeta^{c\bar{c}} \geq \zeta$ for any $\zeta : Y \rightarrow \mathbb{R} \cup \{-\infty\}$.

Then, let us prove $\varphi^{c\bar{c}} = \varphi$ if φ is c -concave. In such a case, we may write

$$\varphi = \zeta^{\bar{c}} \Rightarrow \varphi^c = \zeta^{\bar{c}c} \geq \zeta \Rightarrow \varphi^{c\bar{c}} \leq \zeta^{\bar{c}} = \varphi,$$

where the last inequality is obtained by noting that c - and $\bar{c}$ -transforms revert the inequalities between functions (due to the minus sign in the definition). This proves that in such a case we have $\varphi^{c\bar{c}} \leq \varphi$, and hence $\varphi^{c\bar{c}} = \varphi$.

Finally, we can prove that for any φ , the function $\varphi^{c\bar{c}}$ is the smallest c -concave function larger than φ . To prove that, take $\tilde{\varphi} = \chi^{\bar{c}}$ any c -concave function, and suppose $\tilde{\varphi} \geq \varphi$. Then consider

$$\chi^{\bar{c}} \geq \varphi \Rightarrow \chi^{\bar{c}c} \leq \varphi^c \Rightarrow \chi \leq \varphi^c \Rightarrow \tilde{\varphi} = \chi^{\bar{c}} \geq \varphi^{c\bar{c}},$$

which proves the desired inequality. $\square$ $\square$

We finish this section with a last concept about convex functions, that we will then translate into the framework of c -concave functions later on.

Let us define the graph of the subdifferential of a convex function as

$$\text{Graph}(\partial f) := \{(x, p) : p \in \partial f(x)\} = \{(x, p) : f(x) + f^*(p) = x \cdot p\}.$$

We already know that this graph is monotone in the sense that $(x_i, p_i) \in \text{Graph}(\partial f)$ for $i = 1, 2$ implies

$$(p_2 - p_1) \cdot (x_2 - x_1) \geq 0.$$

⁶Note that this result is easy to prove, since c -concave functions are exactly defined as c -transform of something, while usually convex functions are not defined via sup of affine functions, but via the convexity inequality. This is why the corresponding theorem for convex functions usually requires the use of Hahn-Banach theorem.

Yet, not all monotone graphs are the graphs of the subdifferential of a convex function, nor they are contained in one of such graphs.⁷

Hence, monotonicity is not enough to characterize gradients and subdifferential of convex functions (to be more precise, gradient vector functions that are monotone are gradient of convex functions, but monotonicity alone does not imply *gradientness*).

A stronger notion is that of cyclical monotonicity.

Definition 1.35. A set $A \subset \mathbb{R}^d \times \mathbb{R}^d$ is said to be cyclically monotone if for every $k \in \mathbb{N}$, every permutation σ and every finite family of points $(x_1, p_1), \dots, (x_k, p_k) \in A$ we have

$$\sum_{i=1}^k x_i \cdot p_i \geq \sum_{i=1}^k x_i \cdot p_{\sigma(i)}.$$

The word “cyclical” refers to the fact that, since every σ is the disjoint composition of cycles, it is enough to check this property for cyclical permutations, i.e. replacing $\sum_{i=1}^k x_i \cdot p_{\sigma(i)}$ with $\sum_{i=1}^k x_i \cdot p_{i+1}$ in the definition (with the obvious convention $p_{k+1} = p_1$).

Note that if we take $k = 2$ we get the usual definition of monotonicity, since

$$x_1 \cdot p_1 + x_2 \cdot p_2 \geq x_1 \cdot p_2 + x_2 \cdot p_1 \iff (p_1 - p_2) \cdot (x_1 - x_2) \geq 0.$$

A famous theorem by Rockafellar ([260], Theorem 24.8) states that every cyclically monotone set is contained in the graph of the subdifferential of a convex function. We will not prove this theorem here as will see it as a particular case of a theorem on c -concave functions.

1.6.2 c -cyclical monotonicity and duality

We start from the translation to the c -concave case of the definition of cyclical monotonicity.

Definition 1.36. Once a function $c : \Omega \times \Omega \rightarrow \mathbb{R} \cup \{+\infty\}$ is given, we say that a set $\Gamma \subset \Omega \times \Omega$ is c -cyclically monotone (briefly c -CM) if, for every $k \in \mathbb{N}$, every permutation σ and every finite family of points $(x_1, y_1), \dots, (x_k, y_k) \in \Gamma$ we have

$$\sum_{i=1}^k c(x_i, y_i) \leq \sum_{i=1}^k c(x_i, y_{\sigma(i)}).$$

As for the convex case, the word “cyclical” refers to the fact that we can restrict our attention to cyclical permutations. The word “monotone” is a left-over from the case $c(x, y) = -x \cdot y$.

⁷ Take for instance a 90° rotation R in $\mathbb{R}^2$ and consider the set $A = \{(x, Rx), x \in \mathbb{R}^d\}$. This set satisfies the monotonicity inequality since we have $(Rx_1 - Rx_2) \cdot (x_1 - x_2) = 0$ for any x_1 and x_2 . Yet, the map $x \mapsto Rx$ is not the gradient of a convex functions (it is not a gradient at all), nor can it be contained in the graph of the subdifferential of a convex function.

It is useful to recall the following theorem, which is a generalization of a theorem by Rockafellar in convex analysis. The classical reference for this proof is [269], but it is interesting to see that it can actually be traced back to much older papers, such as [261] (in the framework of rationalizability of actions in economics, Theorem 1 in the paper) or [92] (for applications to liquid crystals, as a part of a proof of a discrete Kantorovich duality for distance costs, see Lemma 2 in the paper).

Theorem 1.37. *If $\Gamma \neq \emptyset$ is a c -CM set in $X \times Y$ and $c : X \times Y \rightarrow \mathbb{R}$ (note that c is required not to take the value $+\infty$), then there exists a c -concave function $\varphi : X \rightarrow \mathbb{R} \cup \{-\infty\}$ (different from the constant $-\infty$ function) such that*

$$\Gamma \subset \{(x, y) \in X \times Y : \varphi(x) + \varphi^c(y) = c(x, y)\}$$

Proof. We will give an explicit formula for the function φ , prove that it is well-defined and that it satisfies the properties that we want to impose.

Let us fix a point $(x_0, y_0) \in \Gamma$: for $x \in X$ set

$$\begin{aligned} \varphi(x) = \inf \{ & c(x, y_n) - c(x_n, y_n) + c(x_n, y_{n-1}) - c(x_{n-1}, y_{n-1}) + \cdots + \\ & + c(x_1, y_0) - c(x_0, y_0) : n \in \mathbb{N}, (x_i, y_i) \in \Gamma \text{ for all } i = 1, \dots, n \}. \end{aligned}$$

Since c is real-valued and Γ is non-empty, φ never takes the value $+\infty$.

If we set, for $y \in Y$,

$$\begin{aligned} -\psi(y) = \inf \{ & -c(x_n, y) + c(x_n, y_{n-1}) - c(x_{n-1}, y_{n-1}) + \cdots + c(x_1, y_0) + \\ & - c(x_0, y_0) : n \in \mathbb{N}, (x_i, y_i) \in \Gamma \text{ for all } i = 1, \dots, n, y_n = y \}. \end{aligned}$$

Note that from the definition we have $\psi(y) > -\infty$ if and only if $y \in (\pi_y)(\Gamma)$. Moreover, by construction we have $\varphi = \psi^c$. This proves that φ is c -concave⁸. The fact that φ is not constantly $-\infty$ can be seen from $\varphi(x_0) \geq 0$: indeed, if we take $x = x_0$, then for any chain of points $(x_i, y_i) \in \Gamma$ we have

$$\sum_{i=0}^n c(x_{i+1}, y_i) \geq \sum_{i=0}^n c(x_i, y_i),$$

where we consider $x_{n+1} = x_0$. This shows that the infimum in the definition of $\varphi(x_0)$ is non-negative.

To prove $\varphi(x) + \varphi^c(y) = c(x, y)$ on Γ it is enough to prove the inequality $\varphi(x) + \varphi^c(y) \geq c(x, y)$ on the same set, since by definition of c -transform the opposite inequality is always true. Moreover, since $\varphi^c = \psi^{\bar{c}c}$ and $\psi^{\bar{c}c} \geq \psi$, it is enough to check $\varphi(x) + \psi(y) \geq c(x, y)$.

Suppose $(x, y) \in \Gamma$ and fix $\varepsilon > 0$. From $\varphi = \psi^c$ one can find a point $\bar{y} \in \pi_y(\Gamma)$ such that $c(x, \bar{y}) - \psi(\bar{y}) < \varphi(x) + \varepsilon$. In particular, $\psi(\bar{y}) \neq \pm\infty$. From the definition of ψ one has the inequality $-\psi(y) \leq -c(x, y) + c(x, \bar{y}) - \psi(\bar{y})$ (since every chain starting from $\bar{y}$ may be completed adding the point $(x, y) \in \Gamma$).

⁸Note that we did not spend words here on the measurability of φ : from its c -concavity, this is straightforward, as it is also continuous, but this is based on the assumption that c is uniformly continuous. Measurability and integrability in the general case are trickier.

Putting together these two informations one gets

$$-\psi(y) \leq -c(x, y) + c(x, \bar{y}) - \psi(\bar{y}) < -c(x, y) + \varphi(x) + \varepsilon,$$

which implies the inequality $c(x, y) \leq \varphi(x) + \psi(y)$ since ε is arbitrary. $\square$ $\square$

We can now prove the following.

Theorem 1.38. *If γ is an optimal transport plan for the cost c and c is continuous, then $\text{spt}(\gamma)$ is a c -CM set.*

Proof. Suppose by contradiction that there exist k , σ and $(x_1, y_1), \dots, (x_k, y_k) \in \text{spt}(\gamma)$ such that

$$\sum_{i=1}^k c(x_i, y_i) > \sum_{i=1}^k c(x_i, y_{\sigma(i)}).$$

Take now $\varepsilon < \frac{1}{2k} (\sum_{i=1}^k c(x_i, y_i) - \sum_{i=1}^k c(x_i, y_{\sigma(i)}))$. By continuity of c , there exists r such that for all $i = 1, \dots, k$ and for all $(x, y) \in B(x_i, r) \times B(y_i, r)$ we have $c(x, y) > c(x_i, y_i) - \varepsilon$ and for all $(x, y) \in B(x_i, r) \times B(y_{\sigma(i)}, r)$ we have $c(x, y) < c(x_i, y_{\sigma(i)}) + \varepsilon$.

Now consider $V_i := B(x_i, r) \times B(y_i, r)$ and note that $\gamma(V_i) > 0$ for every i , because $(x_i, y_i) \in \text{spt}(\gamma)$. Define the measures $\gamma_i := \gamma \llcorner V_i / \gamma(V_i)$ and $\mu_i := (\pi_x)_\# \gamma_i$, $\nu_i := (\pi_y)_\# \gamma_i$. Take $\varepsilon_0 < \frac{1}{k} \min_i \gamma(V_i)$.

For every i , build a measure $\tilde{\gamma}_i \in \Pi(\mu_i, \nu_{\sigma(i)})$ at will (for instance take $\tilde{\gamma}_i = \mu_i \otimes \nu_{\sigma(i)}$).

Now define

$$\tilde{\gamma} := \gamma - \varepsilon_0 \sum_{i=1}^k \gamma_i + \varepsilon_0 \sum_{i=1}^k \tilde{\gamma}_i.$$

We want to find a contradiction by proving that $\tilde{\gamma}$ is a better competitor than γ in the transport problem, i.e. $\tilde{\gamma} \in \Pi(\mu, \nu)$ and $\int c d\tilde{\gamma} < \int c d\gamma$.

First we check that $\tilde{\gamma}$ is a positive measure. It is sufficient to check that $\gamma - \varepsilon_0 \sum_{i=1}^k \gamma_i$ is positive, and, for that, the condition $\varepsilon_0 \gamma_i < \frac{1}{k} \gamma$ will be enough. This condition is satisfied since $\varepsilon_0 \gamma_i = (\varepsilon_0 / \gamma(V_i)) \gamma \llcorner V_i$ and $\varepsilon_0 / \gamma(V_i) \leq \frac{1}{k}$.

Now, let us check the condition on the marginals of $\tilde{\gamma}$. We have

$$\begin{aligned} (\pi_x)_\# \tilde{\gamma} &= \mu - \varepsilon_0 \sum_{i=1}^k (\pi_x)_\# \gamma_i + \varepsilon_0 \sum_{i=1}^k (\pi_x)_\# \tilde{\gamma}_i = \mu - \varepsilon_0 \sum_{i=1}^k \mu_i + \varepsilon_0 \sum_{i=1}^k \mu_i = \mu, \\ (\pi_y)_\# \tilde{\gamma} &= \nu - \varepsilon_0 \sum_{i=1}^k (\pi_y)_\# \gamma_i + \varepsilon_0 \sum_{i=1}^k (\pi_y)_\# \tilde{\gamma}_i = \nu - \varepsilon_0 \sum_{i=1}^k \nu_i + \varepsilon_0 \sum_{i=1}^k \nu_{\sigma(i)} = \nu. \end{aligned}$$

Finally, let us estimate $\int c d\gamma - \int c d\tilde{\gamma}$ and prove that it is positive, thus concluding

the proof. We have

$$\begin{aligned}
\int c \, d\gamma - \int c \, d\tilde{\gamma} &= \varepsilon_0 \sum_{i=1}^k \int c \, d\gamma_i - \varepsilon_0 \sum_{i=1}^k \int c \, d\tilde{\gamma}_i \\
&\geq \varepsilon_0 \sum_{i=1}^k (c(x_i, y_i) - \varepsilon) - \varepsilon_0 \sum_{i=1}^k (c(x_i, y_{\sigma(i)}) + \varepsilon) \\
&= \varepsilon_0 \left(\sum_{i=1}^k c(x_i, y_i) - \sum_{i=1}^k c(x_i, y_{\sigma(i)}) - 2k\varepsilon \right) > 0,
\end{aligned}$$

where we used the fact that γ_i is concentrated on $B(x_i, r) \times B(y_i, r)$, $\tilde{\gamma}_i$ on $B(x_i, r) \times B(y_{\sigma(i)}, r)$, and that they have unit mass (due to the rescaling by $\gamma(V_i)$). $\square$ $\square$

From the previous theorem, together with Theorem 1.37, we get the duality result that we were waiting for⁹ from Section 1.2.

Theorem 1.39. *Suppose that X and Y are Polish spaces and that $c : X \times Y \rightarrow \mathbb{R}$ is uniformly continuous and bounded. Then the problem (DP) admits a solution (φ, φ^c) and we have $\max(\text{DP}) = \min(\text{KP})$.*

Proof. First consider the minimization problem (KP). Since c is continuous, it admits a solution γ . Moreover, the support Γ of γ is c -cyclically monotone, thanks to Theorem 1.38. Then, since c is real valued, we can apply Theorem 1.37 and obtain the existence of a c -concave function φ such that

$$\Gamma \subset \{(x, y) \in \Omega \times \Omega : \varphi(x) + \varphi^c(y) = c(x, y)\}.$$

From their c - (and $\bar{c}$ -) concavity, φ and φ^c are continuous. Moreover, from $\varphi^c(y) = \inf_x c(x, y) - \varphi(x)$ we obtain an upper bound on φ^c (since c is supposed to be bounded), which turns into a lower bound on φ . Symmetrically, we can also obtain upper bounds on φ and lower on φ^c , which prove that φ and φ^c are both continuous and bounded.

Hence, we can use (φ, φ^c) as an admissible pair in (DP). Consider now

$$\int_X \varphi \, d\mu + \int_Y \varphi^c \, d\nu = \int_{X \times Y} (\varphi(x) + \varphi^c(y)) \, d\gamma = \int_{X \times Y} c(x, y) \, d\gamma,$$

where the last equality is due to the fact that γ is concentrated on Γ and there we have $\varphi(x) + \varphi^c(y) = c(x, y)$. The equality just before comes from the fact that the marginals of γ are μ and ν , respectively. This finally shows, using the optimality of γ ,

$$(\text{DP}) \geq \int \varphi \, d\mu + \int \varphi^c \, d\nu = \int c(x, y) \, d\gamma = (\text{KP}),$$

and implies $(\text{DP}) = (\text{KP})$, since we already know $(\text{DP}) \leq (\text{KP})$. As a byproduct of this proof, the pair (φ, φ^c) turns out to be optimal for (DP). $\square$ $\square$

⁹We insist that this is just a possible path: we prove duality via cyclical monotonicity; instead, one could first prove duality (see Section 1.6.3) and then prove cyclical monotonicity through arguments as in Theorem 1.42).

We also want to deal with some cases which are not covered by the previous theorem. First we want to give a short statement about the quadratic case in $\mathbb{R}^d$, in order to comply with Section 1.3.1. The difference compared to Theorem 1.39 lies in the fact that the cost $c(x, y) = \frac{1}{2}|x - y|^2$ is neither bounded, nor uniformly continuous.

Theorem 1.40. *Let μ, ν be probabilities over $\mathbb{R}^d$ and $c(x, y) = \frac{1}{2}|x - y|^2$. Suppose $\int |x|^2 d\mu, \int |y|^2 d\nu < +\infty$. Consider the following variant of (DP)*

$$(\text{DP} - \text{var}) \quad \sup \left\{ \int_{\mathbb{R}^d} \varphi d\mu + \int_{\mathbb{R}^d} \psi d\nu : \varphi \in L^1(\mu), \psi \in L^1(\nu), \varphi \oplus \psi \leq c \right\}.$$

Then (DP-var) admits a solution (φ, ψ) , and the functions $x \mapsto \frac{1}{2}|x|^2 - \varphi(x)$ and $y \mapsto \frac{1}{2}|y|^2 - \psi(y)$ are convex and conjugate to each other for the Legendre transform. Moreover, we have $\max(\text{DP} - \text{var}) = \min(\text{KP})$.

Proof. Applying the same considerations as in Theorem 1.39 we get the existence of a pair (φ, ψ) , with φ c -concave and $\psi = \varphi^c$ such that $\varphi(x) + \varphi^c(y) = c(x, y)$ for every (x, y) in the support of an optimal transport plan γ . From Proposition 1.21 we deduce that $x \mapsto \frac{1}{2}|x|^2 - \varphi(x)$ and $y \mapsto \frac{1}{2}|y|^2 - \psi(y)$ are convex and conjugate to each other. In particular, $\frac{1}{2}|x|^2 - \varphi(x)$ is bounded from below by a linear function, and hence φ is bounded from above by a second-order polynomial. This proves $\varphi_+ \in L^1(\mu)$, from the assumption on μ . The same can be said on ψ , i.e. $\psi_+ \in L^1(\nu)$. We can now integrate $\varphi \oplus \psi$ w.r.t. γ , thus obtaining

$$\int_{\mathbb{R}^d} \varphi d\mu + \int_{\mathbb{R}^d} \psi d\nu = \int_{\mathbb{R}^d \times \mathbb{R}^d} \varphi \oplus \psi d\gamma = \int_{\mathbb{R}^d \times \mathbb{R}^d} c d\gamma \geq 0.$$

This proves $\int_{\mathbb{R}^d} \varphi d\mu, \int_{\mathbb{R}^d} \psi d\nu > -\infty$, and hence $\varphi \in L^1(\mu)$ and $\psi \in L^1(\nu)$.

The conclusion follows as before. $\square$ $\square$

Then, we also give some details about the case where c is not continuous but only l.s.c. We recall that this assumption is sufficient for the existence of an optimal transport plan.

The first result concerns the validity of the duality formula, i.e.

$$\min(\text{KP}) = \sup \left\{ \int_X \varphi d\mu + \int_Y \psi d\nu : \varphi \in C(X), \psi \in C(Y), \varphi \oplus \psi \leq c \right\}. \quad (1.6)$$

By now, we have established this equality when c is uniformly continuous and bounded, also proving that the dual problem admits a maximizing pair. We also know that an inequality is always true : the minimum on the left is always larger than the maximum on the right. More precisely, we are able to deal with the uniformly continuous case (since we want to guarantee continuity of c -concave functions of the form $\varphi(x) = \inf_y c(x, y) - \psi(y)$).

To deal with a l.s.c. cost c bounded from below, we will use the fact there exists a sequence c_k of continuous functions (each one being k -Lipschitz) increasingly converging to c . We need the following lemma.

Lemma 1.41. *Suppose that c_k and c are l.s.c. and bounded from below and that c_k converges increasingly to c . Then*

$$\lim_{k \rightarrow \infty} \min \left\{ \int c_k d\gamma : \gamma \in \Pi(\mu, \nu) \right\} = \min \left\{ \int c d\gamma : \gamma \in \Pi(\mu, \nu) \right\}.$$

Proof. Due to the increasing limit condition, we have $c_k \leq c$ and hence the limit on the left (which exists by monotonicity) is obviously smaller than the quantity on the right. Now consider a sequence $\gamma_k \in \Pi(\mu, \nu)$, built by picking an optimizer for each cost c_k . Up to subsequences, due to the tightness of $\Pi(\mu, \nu)$, we can suppose $\gamma_k \rightharpoonup \bar{\gamma}$. Fix now an index j . Since for $k \geq j$ we have $c_k \geq c_j$, we have

$$\lim_{k \rightarrow \infty} \min \left\{ \int c_k d\gamma : \gamma \in \Pi(\mu, \nu) \right\} = \lim_k \int c_k d\gamma_k \geq \liminf_k \int c_j d\gamma_k.$$

By semi-continuity of the integral cost c_j we have

$$\liminf_k \int c_j d\gamma_k \geq \int c_j d\bar{\gamma}.$$

Hence we have obtained

$$\lim_{k \rightarrow \infty} \min \left\{ \int c_k d\gamma, \gamma \in \Pi(\mu, \nu) \right\} \geq \int c_j d\bar{\gamma}.$$

Since j is arbitrary and $\lim_j \int c_j d\bar{\gamma} = \int c d\bar{\gamma}$ by monotone convergence, we also have

$$\lim_{k \rightarrow \infty} \min \left\{ \int c_k d\gamma, \gamma \in \Pi(\mu, \nu) \right\} \geq \int c d\bar{\gamma} \geq \min \left\{ \int c d\gamma, \gamma \in \Pi(\mu, \nu) \right\}.$$

This concludes the proof. Note that it also gives, as a byproduct, the optimality of $\bar{\gamma}$ for the limit cost c . $\square$ $\square$

We can now establish the validity of the duality formula for semi-continuous costs.

Theorem 1.42. *If X, Y are Polish spaces and $c : X \times Y \rightarrow \mathbb{R} \cup \{+\infty\}$ is l.s.c. and bounded from below, then the duality formula $\min(\text{KP}) = \sup(\text{DP})$ holds.*

Proof. Consider a sequence c_k of k -Lipschitz functions approaching c increasingly. Then the same duality formula holds for c_k , and hence we have

$$\begin{aligned} \min \left\{ \int c_k d\gamma, \gamma \in \Pi(\mu, \nu) \right\} &= \max \left\{ \int \varphi d\mu + \int \psi d\nu : \varphi \oplus \psi \leq c_k \right\} \\ &\leq \sup \left\{ \int \varphi d\mu + \int \psi d\nu : \varphi \oplus \psi \leq c \right\}, \end{aligned}$$

where the max and the sup are computed among C_b functions φ, ψ . The inequality is justified by the fact that $c_k \leq c$ and hence every pair (φ, ψ) satisfying $\varphi(x) + \psi(y) \leq c_k(x, y)$ also satisfies $\varphi(x) + \psi(y) \leq c(x, y)$. The conclusion follows by letting $k \rightarrow +\infty$, using Lemma 1.41. Note that for the cost c we cannot guarantee the existence of a maximizing pair (φ, ψ) . $\square$ $\square$

The duality formula also allows to prove the following c -cyclical monotonicity theorem.

Theorem 1.43. *If c is l.s.c. and γ is an optimal transport plan, then γ is concentrated on a c -CM set Γ (which will not be closed in general).*

Proof. Thanks to the previous theorem the duality formula holds. This means that, if we take a sequence of maximizing pairs (φ_h, ψ_h) in the dual problem, we have

$$\int (\varphi_h(x) + \psi_h(y)) d\gamma = \int \varphi_h d\mu + \int \psi_h d\nu \rightarrow \int c d\gamma,$$

since the value of $\int c d\gamma$ is the minimum of the primal problem, which is also the maximum of the dual. Moreover, we have $c(x, y) - \varphi_h(x) + \psi_h(y) \geq 0$, which implies that the functions $f_h := c(x, y) - \varphi_h(x) - \psi_h(y)$, defined on $X \times Y$, converge to 0 in $L^1(X \times Y, \gamma)$ (since they are positive and their integral tends to 0). As a consequence, up to a subsequence (not relabeled) they also converge pointwisely γ -a.e. to 0. Let $\Gamma \subset X \times Y$ be a set with $\gamma(\Gamma) = 1$ where the convergence holds.

Take any k, σ and $(x_1, y_1), \dots, (x_k, y_k) \in \Gamma$: we have

$$\sum_{i=1}^k c(x_i, y_i) = \lim_h \sum_{i=1}^k \varphi_h(x_i) + \psi_h(y_i) = \lim_h \sum_{i=1}^k \varphi_h(x_i) + \psi_h(y_{\sigma(i)}) \leq \sum_{i=1}^k c(x_i, y_{\sigma(i)}),$$

which proves that this set is c -CM. □ □

Remark 1.44. The duality formula that we proved for l.s.c. costs c differs from that for continuous costs in that there is no existence for the dual problem (DP). Actually, if one restates the dual problem as in (DP-var), then one can produce an optimizer by first applying Theorem 1.43 to say that the optimal γ is concentrated on a c -CM set Γ , and then build a potential φ through Theorem 1.37. This works under the assumption that c is real-valued, and does not depend on its continuity¹⁰.

1.6.3 A direct proof of duality

In section 1.6.2 we paid off the debt that we contracted with duality in Section 1.2: we proved, via c -cyclical monotonicity arguments, that $\min(\text{KP}) = \sup(\text{DP})$, with no duality gap, under some very general assumptions.

Here we want to give an alternative proof, independent of c -cyclical monotonicity and based on a simple convex analysis trick¹¹. We will restrict to the compact case, and only use the following fact: if H is convex and l.s.c., then $H^{**} = H$ (see Box 1.12 in Section 1.6.1).

¹⁰Yet, note that the construction requires to use an optimal γ , which is guaranteed in the case where c is l.s.c., but could be the case in some special situations as well, such as when we can apply Lemma 1.8. Also, for non-uniformly continuous costs c , there are measurability issues on the function φ of Theorem 1.37.

¹¹Even if not strictly necessary, we prefer to give it for completeness, and because it is different than what is presented in other texts, such as [292], where the main tool is a theorem from [91], Chapter 1, applied to well-chosen functions and space. The present proof, suggested by C. Jimenez, is essentially adapted from Section 4 in [71].

Suppose that X and Y are compact metric spaces, and $c : X \times Y \rightarrow \mathbb{R}$ is continuous. For every $p \in C(X \times Y)$ define

$$H(p) := -\max \left\{ \int_X \varphi d\mu + \int_Y \psi d\nu : \varphi(x) + \psi(y) \leq c(x, y) - p(x, y) \right\},$$

which coincides with the opposite of the value of (DP) with cost $c - p$. The fact that this value is attained is proven in Proposition 1.11, which also provides bounds on the modulus of continuity of the optimal φ and ψ (the same as $c - p$) and on their oscillation (if we assume $\varphi(x_0) = 0$, then both φ and ψ are bounded by a constant only depending on $\sup |c - p|$).

Lemma 1.45. *The function $H : C(X \times Y) \rightarrow \mathbb{R}$ is convex and l.s.c. for the uniform convergence on the compact space $X \times Y$.*

Proof. For convexity, take p_0 and p_1 with their optimal potentials (φ_0, ψ_0) and (φ_1, ψ_1) . For $t \in [0, 1]$ define $p_t = (1-t)p_0 + t p_1$, $\varphi_t = (1-t)\varphi_0 + t\varphi_1$, and $\psi_t = (1-t)\psi_0 + t\psi_1$. The pair (φ_t, ψ_t) is admissible in the max defining $-H(p_t)$. We have

$$H(p_t) \leq - \left(\int_X \varphi_t d\mu + \int_Y \psi_t d\nu \right) = (1-t)H(p_0) + tH(p_1),$$

which shows convexity.

For semi-continuity, take $p_n \rightarrow p$. Extract a subsequence p_{n_k} realizing the liminf of $H(p_n)$. From uniform convergence, the sequence p_{n_k} is equicontinuous and bounded (use the converse of the Ascoli Arzelà Theorem). Hence, the corresponding optimal potential $(\varphi_{n_k}, \psi_{n_k})$ are also equicontinuous and bounded and we can assume $\varphi_{n_k} \rightarrow \varphi$, $\psi_{n_k} \rightarrow \psi$ uniformly, up to extracting again. Obviously $\varphi_{n_k}(x) + \psi_{n_k}(y) \leq c(x, y) - p_{n_k}(x, y)$ implies $\varphi(x) + \psi(y) \leq c(x, y) - p(x, y)$. Hence

$$H(p) \leq - \left(\int_X \varphi d\mu + \int_Y \psi d\nu \right) = \lim_k H(p_{n_k}) = \liminf_n H(p_n),$$

which proves lower semi-continuity. □ □

Let us now compute $H^* : \mathcal{M}(X \times Y) \rightarrow \mathbb{R} \cup \{+\infty\}$. For $\gamma \in \mathcal{M}(X \times Y)$ we have

$$H^*(\gamma) := \sup_p \int_{X \times Y} p d\gamma + \sup \left\{ \int_X \varphi d\mu + \int_Y \psi d\nu : \varphi(x) + \psi(y) \leq c(x, y) - p(x, y) \right\},$$

which can be written as a unique sup over p, φ, ψ . Note that, if $\gamma \notin \mathcal{M}_+(X \times Y)$, then there is $p \leq 0$ such that $\int p_0 d\gamma > 0$ and one can take $\varphi = 0$, $\psi = 0$, $p = c + n p_0$ and, for $n \rightarrow \infty$, we get $H^*(\gamma) = +\infty$. On the contrary, if $\gamma \in \mathcal{M}_+(X \times Y)$, we should choose the largest possible p , i.e. $p(x, y) = c(x, y) - \varphi(x) - \psi(y)$. This gives

$$H^*(\gamma) = \sup_{\varphi, \psi} \int_{X \times Y} c(x, y) d\gamma + \int_X \varphi d\mu - \int_{X \times Y} \varphi(x) d\gamma + \int_Y \psi d\nu - \int_{X \times Y} \psi(y) d\gamma.$$

But this is exactly the expression that we introduced in Section 1.2 (see equation (1.3)) to re-write the constraints in (KP)! Indeed we have

$$H^*(\gamma) = \begin{cases} K(\gamma) & \text{if } \gamma \in \Pi(\mu, \nu), \\ +\infty & \text{if not.} \end{cases}$$

The duality theorem follows.

Theorem 1.46. *If X, Y are compact spaces and c continuous, then $\min(\text{KP}) = \sup(\text{DP})$ (proof by convex analysis).*

Proof. We have $\max(\text{DP}) = -H(0) = -H^{**}(0)$ (since H is convex and l.s.c., see Box 1.12). Moreover, $H^{**}(0) = \sup_{\gamma} \langle 0, \gamma \rangle - H^*(\gamma) = -\inf H^* = -\min(\text{KP})$. $\square$ $\square$

1.6.4 Sufficient conditions for optimality and stability

The considerations of Sections 1.6.1 and 1.6.2 about c -transforms allow us to prove sufficient conditions for optimality in optimal transportation, at least when the cost satisfies the twist condition of Definition 1.16. As we did in Section 1.3, we will first a general statement under compactness assumptions, and we will then give a different proof in the quadratic case, with no need of compactness.

Theorem 1.47. *Let $\Omega \subset \mathbb{R}^d$ be compact, and c be a C^1 cost function satisfying the twist condition on $\Omega \times \Omega$. Suppose that $\mu \in \mathcal{P}(\Omega)$ and $\varphi \in c\text{-conc}(\Omega)$ are given, that φ is differentiable μ -a.e. and that $\mu(\partial\Omega) = 0$. Suppose that the map T satisfies $\nabla_x c(x, T(x)) = \nabla \varphi(x)$. Then T is optimal for the transport cost c between the measures μ and $\nu := T_{\#}\mu$.*

Proof. Consider the function φ , which may be written as $\varphi(x) = \inf_y c(x, y) - \psi(y)$ for a certain function $\psi : \Omega \rightarrow \mathbb{R}$. The function ψ may be supposed to be $\bar{c}$ -concave and hence continuous (actually, we can take $\psi = \varphi^c$, from the considerations of the previous section). Fix now $x_0 \in \Omega$ such that $\nabla \varphi(x_0)$ exists and $x_0 \notin \partial\Omega$. By compactness and continuity, one can say that $\inf_y c(x, y) - \psi(y)$ is realized by a certain point y_0 . This gives

$$\varphi(x) \leq h(x - y_0) - \psi(y_0) \quad \text{for every } x \quad \varphi(x_0) = h(x_0 - y_0) - \psi(y_0),$$

and hence $x \mapsto h(x - y_0) - \varphi(x)$ is minimal at $x = x_0$ (note that we defined y_0 by the optimality in y but now we use the optimality in x). As a consequence, we get

$$\nabla \varphi(x_0) = \nabla_x c(x_0, y_0).$$

By assumption $y \mapsto \nabla_x c(x_0, y)$ is injective: this implies $y_0 = T(x_0)$. This proves $\varphi(x_0) + \psi(T(x_0)) = c(x_0, T(x_0))$ and this same equality is true for μ -a.e. x_0 . If we integrate with respect to μ we get

$$\int \varphi \, d\mu + \int \psi \, d\nu = \int \varphi \, d\mu + \int \psi \circ T \, d\mu = \int c(x, T(x)) \, d\mu(x),$$

which proves the optimality of T since the last integral equals the cost of T in the problem of Monge, and we have $\int \varphi \, d\mu + \int \psi \, d\nu \leq (\text{DP}) \leq (\text{KP})$, since (φ, ψ) is admissible in (DP). $\square$ $\square$

As we promised, here is a different proof for the case $c(x, y) = \frac{1}{2}|x - y|^2$.

Theorem 1.48. *Suppose that $\mu \in \mathcal{P}(\mathbb{R}^d)$ is such that $\int |x|^2 d\mu(x) < \infty$, that $u : \mathbb{R}^d \rightarrow \mathbb{R} \cup \{+\infty\}$ is convex and differentiable μ -a.e. Set $T = \nabla u$ and suppose $\int |T(x)|^2 d\mu(x) < +\infty$. Then T is optimal for the transport cost $c(x, y) := \frac{1}{2}|x - y|^2$ between the measures μ and $\nu := T_{\#}\mu$.*

Proof. First note that, for a convex function u , we have the following properties

$$u(x) + u^*(y) \geq x \cdot y \text{ for all } x, y \in \mathbb{R}^d, \quad u(x) + u^*(y) = x \cdot y \text{ if } y = \nabla u(x).$$

Now, consider any transport plan $\gamma \in \Pi(\mu, \nu)$ and write

$$\begin{aligned} \int_{\mathbb{R}^d \times \mathbb{R}^d} (x \cdot y) d\gamma(x, y) &\leq \int_{\mathbb{R}^d \times \mathbb{R}^d} (u(x) + u^*(y)) d\gamma(x, y) \\ &= \int_{\mathbb{R}^d} u(x) d\mu(x) + \int_{\mathbb{R}^d} u^*(T(x)) d\mu(x) = \int_{\mathbb{R}^d} (x \cdot T(x)) d\mu(x). \end{aligned}$$

This proves $\int (x \cdot y) d\gamma \leq \int (x \cdot y) d\gamma_T$. Subtracting these quantities from the integral, which only depends on μ and ν , $\int \frac{1}{2}(|x|^2 + |y|^2) d\gamma = \int \frac{1}{2}(|x|^2 + |y|^2) d\gamma_T$ we get

$$\int_{\mathbb{R}^d \times \mathbb{R}^d} \frac{1}{2}|x - y|^2 d\gamma \geq \int_{\mathbb{R}^d \times \mathbb{R}^d} \frac{1}{2}|x - y|^2 d\gamma_T,$$

which proves the claim. $\square$ $\square$

Let us spend some words on the two different proofs that we just gave: we needed to prove some optimality properties of the point $T(x)$ among possible points y . In the first case, we proved that an optimal point y_0 existed, and we deduced from the first order optimality conditions $y_0 = T(x_0)$. In the second case, we used the characterization of minimizers for convex functions: whenever a gradient vanishes, we are at a minimizer. This is the idea behind the property $u(x) + u^*(y) = x \cdot y$ if $y = \nabla u(x)$.

We now pass to a more general criterion that can be used for almost arbitrary costs. The main idea is that we proved that optimal plans are necessarily concentrated on c -CM sets, but the converse is also true: a plan which is concentrated on a c -CM set is optimal, at least in the most reasonable cases. We will prove this fact in the easiest case, i.e. when c is uniformly continuous and bounded. Actually, we will see in the next proof that the main ingredient is Theorem (1.37), which only requires finiteness of the cost, but continuity and compactness are needed to avoid integrability issues (in this case all the functions are measurable and bounded). Moreover, we will apply this sufficient criterion to another interesting problem, i.e. stability of the optimal plans, which will require these assumptions for other reasons. The fact that this converse implication stays true when these assumptions are withdrawn (and in particular for infinite costs) is a delicate matter, and we refer to [11] for a counter-example when the cost takes the value $+\infty$ and to [254, 31, 62] for interesting positive proofs.

Theorem 1.49. *Suppose that $\gamma \in \mathcal{P}(X \times Y)$ is given, that X and Y are Polish spaces, that $c : X \times Y \rightarrow \mathbb{R}$ is uniformly continuous and bounded, and that $\text{spt}(\gamma)$ is c -CM. Then γ is an optimal transport plan between its marginals $\mu = (\pi_x)_{\#}\gamma$ and $\nu = (\pi_y)_{\#}\gamma$ for the cost c .*

Proof. Theorem (1.37) gives the existence of a c -concave function φ such that $\text{spt}(\gamma)$ is contained in the set $\{(x, y) : \varphi(x) + \varphi^c(y) = c(x, y)\}$. Both φ and φ^c are continuous thanks to the continuity of c , and hence bounded on X and Y , respectively.

Thanks to what we know about duality we have

$$(\text{KP}) \leq \int c(x, y) d\gamma = \int (\varphi(x) + \varphi^c(y)) d\gamma = \int \varphi d\mu + \int \varphi^c d\nu \leq (\text{DP}) = (\text{KP})$$

which shows that γ is optimal (and that (φ, φ^c) solves the dual problem). $\square$ $\square$

We are now able to prove the following stability result that we only state in the compact case.

Theorem 1.50. *Suppose that X and Y are compact metric spaces and that $c : X \times Y \rightarrow \mathbb{R}$ is continuous. Suppose that $\gamma_n \in \mathcal{P}(X \times Y)$ is a sequence of transport plan which are optimal for the cost c between their own marginals $\mu_n := (\pi_x)_\# \gamma_n$ and $\nu_n := (\pi_y)_\# \gamma_n$, and suppose $\gamma_n \rightarrow \gamma$. Then $\mu_n \rightarrow \mu := (\pi_x)_\# \gamma$, $\nu_n \rightarrow \nu := (\pi_y)_\# \gamma$ and γ is optimal in the transport between μ and ν .*

Proof. Set $\Gamma_n := \text{spt}(\gamma_n)$. Up to subsequences, we can assume $\Gamma_n \rightarrow \Gamma$ in the Hausdorff topology (see below). Each support Γ_n is a c -CM set (Theorem (1.38)) and the Hausdorff limit of c -CM sets is also c -CM. Indeed, if one fixes $(x_1, y_1), \dots, (x_k, y_k) \in \Gamma$ there are points $(x_1^n, y_1^n), \dots, (x_k^n, y_k^n) \in \Gamma_n$ such that, for each $i = 1, \dots, k$, we have $(x_i^n, y_i^n) \rightarrow (x_i, y_i)$. The cyclical monotonicity of Γ_n gives $\sum_{i=1}^k c(x_i^n, y_i^n) \leq \sum_{i=1}^k c(x_i^n, y_{\sigma(i)}^n)$, which implies, taking the limit $n \rightarrow \infty$

$$\sum_{i=1}^k c(x_i, y_i) \leq \sum_{i=1}^k c(x_i, y_{\sigma(i)}).$$

This proves that Γ is c -CM. Moreover, we know from the convergence $\gamma_n \rightarrow \gamma$ together with $\Gamma_n \rightarrow \Gamma$, that $\text{spt}(\gamma) \subset \Gamma$. This shows that $\text{spt}(\gamma)$ is c -CM and implies the optimality of γ $\square$ $\square$

We finish with an easy but useful consequence of Theorem 1.50. To fix the notations, for a given cost $c : X \times Y \rightarrow \mathbb{R}$ and $\mu \in \mathcal{P}(X)$, $\nu \in \mathcal{P}(Y)$, let us define

$$\mathcal{T}_c(\mu, \nu) := \min \left\{ \int_{X \times Y} c d\gamma : \gamma \in \Pi(\mu, \nu) \right\}.$$

Theorem 1.51. *Suppose that X and Y are compact metric spaces and that $c : X \times Y \rightarrow \mathbb{R}$ is continuous. Suppose that $\mu_n \in \mathcal{P}(X)$ and $\nu_n \in \mathcal{P}(Y)$ are two sequences of probability measures, with $\mu_n \rightarrow \mu$ and $\nu_n \rightarrow \nu$. Then we have $\mathcal{T}_c(\mu_n, \nu_n) \rightarrow \mathcal{T}_c(\mu, \nu)$.*

Proof. Let γ_n be an optimal transport plan from μ_n to ν_n for the cost c . Up to subsequences, we can assume $\gamma_n \rightarrow \gamma$. Theorem 1.50 provides the optimality of γ . This means that we have (along this subsequence, but since it is arbitrary...)

$$\mathcal{T}_c(\mu_n, \nu_n) = \int_{X \times Y} c d\gamma_n \rightarrow \int_{X \times Y} c d\gamma = \mathcal{T}_c(\mu, \nu)$$

and proves the claim. $\square$ $\square$

Box 1.13. – Memo – Hausdorff convergence

Definition - In a compact metric space X we define the Hausdorff distance on pair of compact subsets of X by setting

$$d_H(A, B) := \max \{ \max \{ d(x, A) : x \in B \}, \max \{ d(x, B) : x \in A \} \}.$$

Properties - We have the following equivalent definition:

- 1) $d_H(A, B) = \max \{ |d(x, A) - d(x, B)| : x \in X \}$
- 2) $d_H(A, B) = \inf \{ \varepsilon > 0 : A \subset B_\varepsilon, B \subset A_\varepsilon \}$, where A_ε and B_ε stand for the ε -neighborhood of A and B , respectively.

Theorem (Blaschke) - d_H is a distance: it is positive and symmetric, it only vanishes if the two sets coincide, and satisfies triangle inequality. With this distance, the set of compact subsets of X becomes a compact metric space itself.

We refer to [13] for a detailed discussion (with proofs) of this topic. Here we only prove a simple fact that we need.

Proposition - If $d_H(A_n, A) \rightarrow 0$ and μ_n is a sequence of positive measures such that $\text{spt}(\mu_n) \subset A_n$ with $\mu_n \rightarrow \mu$, then $\text{spt}(\mu) \subset A$.

Sketch of proof - For each n we have $\int d(x, A_n) d\mu_n = 0$, since μ_n is supported on A_n . Since $d(x, A_n) \rightarrow d(x, A)$ uniformly and $\mu_n \rightarrow \mu$, thanks to the duality between uniform convergence and weak convergence of measures we get $\int d(x, A_n) d\mu_n \rightarrow \int d(x, A) d\mu$. This implies $\int d(x, A) d\mu = 0$ and hence μ is concentrated on A .

Let us also conclude with a result on the stability of the Kantorovich potentials.

Theorem 1.52. *Suppose that X and Y are compact metric spaces and that $c : X \times Y \rightarrow \mathbb{R}$ is continuous. Suppose that $\mu_n \in \mathcal{P}(X)$ and $\nu_n \in \mathcal{P}(Y)$ are two sequences of probability measures, with $\mu_n \rightarrow \mu$ and $\nu_n \rightarrow \nu$. Let (φ_n, ψ_n) be, for each n , a pair of c -concave Kantorovich potentials for the cost c in the transport from μ_n to ν_n . Then, up to subsequences, we have $\varphi_n \rightarrow \varphi$, $\psi_n \rightarrow \psi$, where the convergence is uniform and (φ, ψ) is a pair of Kantorovich potentials for μ and ν .*

Proof. We know that c -concave functions have the same modulus of continuity as c and hence, up to translating by a constant, we can apply the Ascoli Arzelà theorem. We have $\varphi_n \rightarrow \tilde{\varphi}$ and $\psi_n \rightarrow \tilde{\psi}$, the convergences being uniform. From $\varphi_n(x) + \psi_n(y) \leq c(x, y)$ we deduce, as $n \rightarrow \infty$, the inequality $\tilde{\varphi}(x) + \tilde{\psi}(y) \leq c(x, y)$. Moreover we have

$$\mathcal{T}_c(\mu_n, \nu_n) = \int \varphi_n d\mu_n + \int \psi_n d\nu_n \rightarrow \int \tilde{\varphi} d\mu + \int \tilde{\psi} d\nu.$$

Yet, we also have (Theorem 1.50) $\mathcal{T}_c(\mu_n, \nu_n) \rightarrow \mathcal{T}_c(\mu, \nu)$. We infer that the pair $(\tilde{\varphi}, \tilde{\psi})$ is admissible in the dual problem and realizes the maximal value $\int \tilde{\varphi} d\mu + \int \tilde{\psi} d\nu = \mathcal{T}_c(\mu, \nu)$. Hence they are Kantorovich potentials.

The convergence is obviously true on the full sequence (without passing to a subsequence) in case of uniqueness of the Kantorovich potentials at the limit. $\square \quad \square$

1.7 Discussion

This section presents alternative point of views (probabilistic language, polar factorization), connections with economics and finance, generalizations and a brief presentation of the regularity issues on the optimal transport map and the optimal potentials.

1.7.1 Probabilistic interpretation

We completely skipped up to now any interpretations involving probabilities, which are actually natural in this setting. By changing for a while our language, let us describe the situation in the following way.

Two probability measures μ and ν on some spaces (often, $\mathbb{R}^d$) are given, and can be interpreted as the laws of two random variables. Yet, we do not prescribe the joint law (which corresponds to γ) of these two random variables and we consider the optimization problem

$$\min\{\mathbb{E}[c(X, Y)] : X \sim \mu, Y \sim \nu\}$$

where $\mathbb{E}$ denotes the expected value (according to a probability $\mathbb{P}$ on a probability space Ω , which is not relevant for the minimization, but could be considered either “big enough” or to be part of the optimization). This expected value obviously depends on the joint law of (X, Y) , which is the main unknown.

The particular case of real (or vector) valued r.v. with $c(X, Y) = |X - Y|^p$ reads

$$\min\{\|X - Y\|_{L^p} : X \sim \mu, Y \sim \nu\}.$$

More interesting is the case $p = 2$, where the problem can be expressed in terms of covariance. Indeed, let us set $x_0 = \mathbb{E}[X] = \int x \, d\mu$ and $y_0 = \mathbb{E}[Y] = \int y \, d\nu$, these two values being the mean values of X and Y . We have

$$\begin{aligned} \mathbb{E}[|X - Y|^2] &= \mathbb{E}[|(X - x_0) - (Y - y_0) + (x_0 - y_0)|^2] \\ &= \mathbb{E}[|X - x_0|^2] + \mathbb{E}[|Y - y_0|^2] + |x_0 - y_0|^2 \\ &\quad + 2\mathbb{E}[X - x_0] \cdot (x_0 - y_0) - 2\mathbb{E}[Y - y_0] \cdot (x_0 - y_0) \\ &\quad - 2\mathbb{E}[(X - x_0) \cdot (Y - y_0)]. \end{aligned}$$

In this expression, the three first terms only depend on the laws of X and Y separately (the first being the variance of X , the second the variance of Y , and the third the squared distance between the two mean values), and the next two terms vanish (since the mean value of $X - x_0$ is 0, and so for the mean value of $Y - y_0$). The problem is hence reduced to the maximization of $\mathbb{E}[(X - x_0) \cdot (Y - y_0)]$. This means that we need to find the joint law which guarantees maximal covariance (i.e. somehow maximal dependence) of two r.v. with given laws. In the case of real valued r.v. (see next chapter) the answer will be that the optimal coupling is obtained in the case where X and Y are completely dependent, and one is an increasing function of the other. The multidimensional case obviously replaces this increasing behavior with other monotone behaviors.